23 Ans: A

Magnetic flux density due to horizontal wire $= \dfrac{\mu_o I}{2\pi d}$

$= \dfrac{(4\pi \times 10^{-7})(7.00)}{2\pi(3.00)}$

$= 4.67 \times 10^{-7}$ T, out of the page

Magnetic flux density due to vertical wire $= \dfrac{\mu_o I}{2\pi d}$

$= \dfrac{(4\pi \times 10^{-7})(6.00)}{2\pi(4.00)}$

$= 3.00 \times 10^{-7}$ T, into the page

Resultant B = 1.67×10^{-7} T out of the page.

24 Ans: D